Business Report

Statistical Analysis and Modelling

PGPDSGA

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1. Context

We have a set of business problems. We need to find solutions to these problems, arrive at the data insights and suggest to business with suggestions and insights to help the business to become successful and drive business outcome.

2. Objective

The goal is to find relevant relationship between different business entities and items and do statistical analysis and perform required statistical tests and check relevance and significance of your hypothesis. This will help to derive insights.

3. Problem Statement

These are the problem statements identified from multiple companies/ people:

A physiotherapist for a male football team is investigating the relationship between player positions and foot injuries. The goal is to calculate the probabilities of injuries occurring across various field positions (Striker, Forward, Attacking Midfielder, and Winger) based on historical data.

The quality team of the cement company wants to know about the packaging material to better understand wastage or pilferage within the supply chain. We need to calculate proportion of the gunny bags have a particular breaking strength or less than or greater than a particular breaking strength

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. They need to know whether it is good for printing, and what is hardeness difference between polished and unpolished stones.

The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest. Company wants to understand how hardness of implants vary depending on dentists or methods either separately or together, as well as its variability based on alloy.

4. Problem 1: Football Injury Analysis

4.1. Context and Objective

A physiotherapist for a male football team is investigating the relationship between player positions and foot injuries. The goal is to calculate the probabilities of injuries occurring across various field positions (Striker, Forward, Attacking Midfielder, and Winger) based on historical data.

4.2. Descriptive Statistics (Contingency Table)

Table 1 Contingency table

Position	**Players Injured**	**Players Not Injured**	**Total**
Striker	45	32	77
Forward	56	38	94
Attacking Midfielder	24	11	35
Winger	20	9	29
Total	**145**	**90**	**235**

4.3. Probability Analysis

Based on the contingency table, the following probabilities have been calculated:

Question1.1. Probability of a player suffering an injury:

$$P(\text{Injured}) = \frac{\text{Total Injured}}{\text{Grand Total}} = \frac{145}{235} \approx 0.6170$$

The overall probability of a player sustaining an injury is approximately 61.7%.

Probability that a randomly chosen player would suffer an injury: 0.617

Question1.2. Probability that a player is a Forward or a Winger:

$$P(\text{Forward} \cup \text{Winger}) = \frac{94 + 29}{235} = \frac{123}{235} \approx 0.5234$$

There is a 52.34% chance that a randomly chosen player is either a forward or a winger.

Probability that a player is a forward or a winger: 0.5234

Question1.3. Probability of being a Striker AND having a foot injury:

$$P(\text{Striker} \cap \text{Injured}) = \frac{45}{235} \approx 0.1915$$

The joint probability of a player being a striker and having an injury is 19.15%.

Probability that a randomly chosen player plays in a striker position and has a foot injury: 0.1915

Question1.4. Probability that an injured player is a Striker:

$$P(\text{Striker}|\text{Injured}) = \frac{45}{145} \approx 0.3103$$

Given that a player is injured, there is a 31.03% probability that they are a striker.

Probability that a randomly chosen injured player is a striker: 0.3103

All the analysis for Problem 1 is complete and business answers are given above.

---

5. Problem 2: Packaging Material Strength Analysis

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information; (Provide an appropriate visual representation of your answers, without which marks will be deducted)

2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?

5.1. Context and Objective

The quality team of a cement company is analyzing the breaking strength of gunny bags to minimize wastage. The breaking strength follows a **Normal Distribution** with a Mean of 5 kg/sq cm and a Standard Deviation of 1.5 kg/sq cm.

5.2. Normal Distribution Analysis

Question2.1. Proportion with breaking strength < 3.17 kg/sq cm:

Mean (mu): 5 kg/sq cm

Standard Deviation (sigma): 1.5 kg/sq cm

Value (x): 3.17 kg/sq cm

Calculate the Z-score

z = (x - mu) / sigma

$$Z = \frac{3.17-5}{1.5} = -1.22$$

Z-score: -1.2200

Calculate the proportion (cumulative probability) for less than x, using Cumulative Distribution Function :- proportion = stats.norm.cdf(x, loc=mu, scale=sigma)

$$P(X < 3.17) = P(Z < -1.22) \approx \mathbf{0.1112}$$

Proportion of gunny bags with breaking strength less than 3.17 kg/sq cm: 0.1112

Z-score: -1.2200

Question2.1 : Answer:

Proportion of gunny bags with breaking strength less than 3.17 kg/sq cm: 0.1112

Visual Representation:

The shaded portion shows the probability

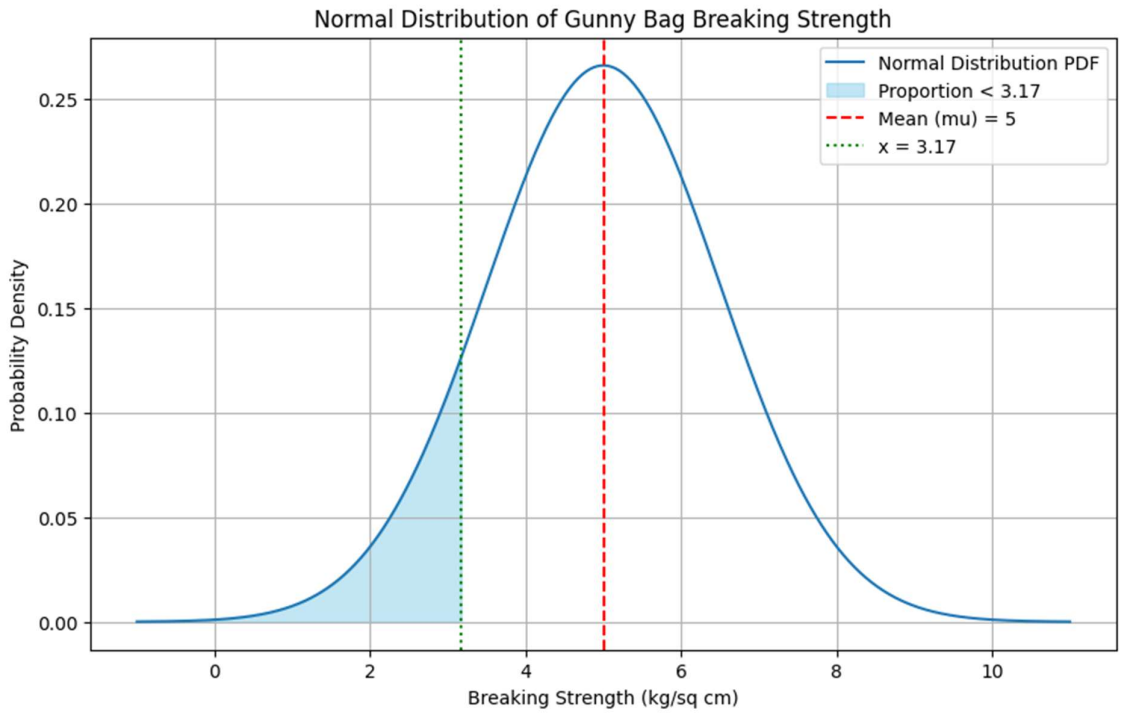

Figure 1: Proportion of gunny bags with breaking strength less than 3.17 kg/sq cm

Question 2.2. Proportion with breaking strength of at least 3.6 kg/sq cm:

```
mu = 5  # Mean breaking strength (kg/sq cm)

sigma = 1.5  # Standard deviation (kg/sq cm)

x = 3.6  # Breaking strength value
```

```
Calculate the Z-score

z = (x - mu) / sigma
```

```
Z-score: -0.9333
```

Calculate **1 - proportion_less_than_x** , Using the Cumulative Distribution Function (CDF)

which is 1- stats.norm.cdf(x, loc=mu, scale=sigma)

$$Z = \frac{3.6-5}{1.5} = -0.9333$$

$$P(X \geq 3.6) = 1 - P(Z < -0.9333) \approx \mathbf{0.8247}$$

Question 2.2: Answer :

Proportion of gunny bags with breaking strength at least 3.6 kg/sq cm: 0.8247

Visual Representation:

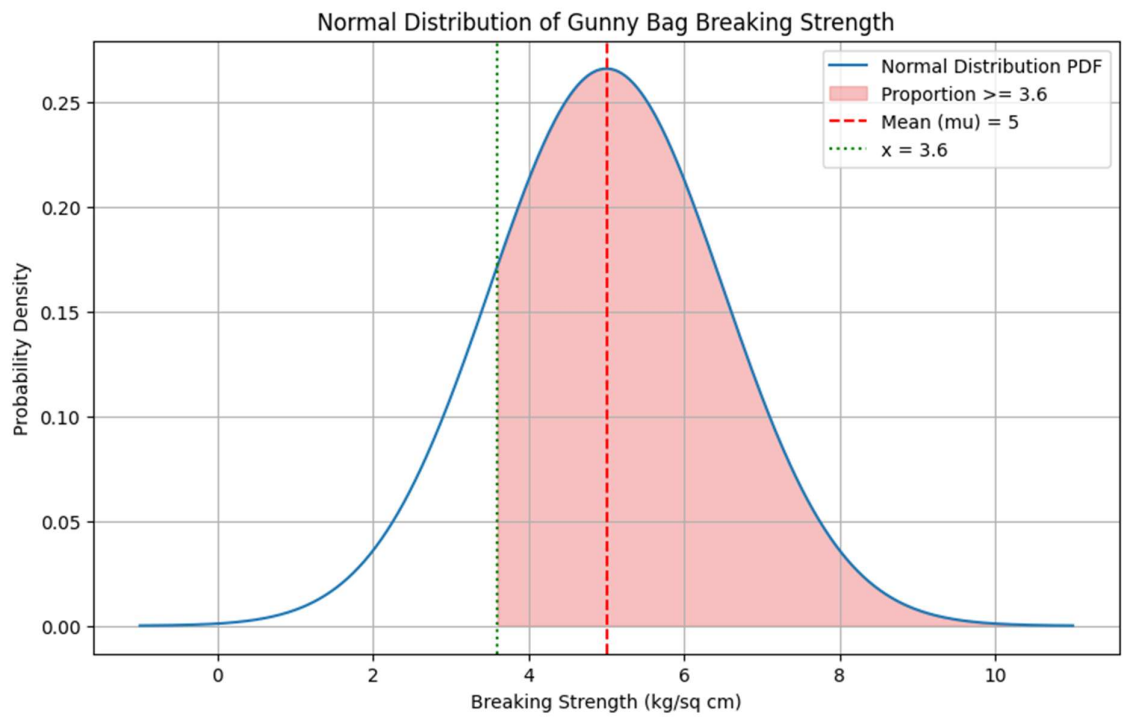

Figure 2:Proportion of gunny bags with breaking strength at least 3.6 kg/sq cm

Question 2.3. Proportion with breaking strength between 5 and 5.5 kg/sq cm:

```
mu = 5  # Mean breaking strength (kg/sq cm)

sigma = 1.5  # Standard deviation (kg/sq cm)

x1 = 5  # Lower bound

x2 = 5.5 # Upper bound
```

Calculate the Z-scores for x1 and x2

z1 = (x1 - mu) / sigma

z2 = (x2 - mu) / sigma

Calculate the CDF (Cumulative Distribution Frequency) proportion between cumulative probability for x1 and x2

stats.norm.cdf(x2, loc=mu, scale=sigma) minus stats.norm.cdf(x1, loc=mu, scale=sigma)

Values:-

Value (x1): 5 kg/sq cm, Z-score: 0.0000

Value (x2): 5.5 kg/sq cm, Z-score: 0.3333

Question2.3 Answer:

Answer: Proportion of gunny bags with breaking strength between 5 and 5.5 kg/sq cm: 0.1306

Visual Representation:

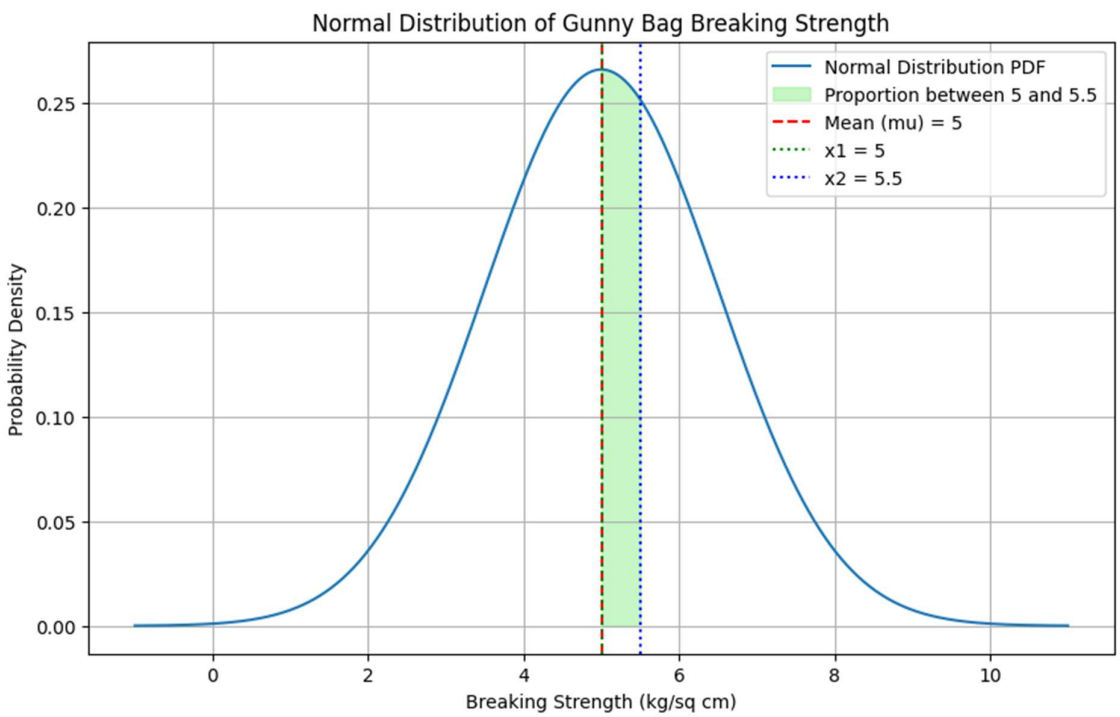

Figure 3: Proportion of gunny bags with breaking strength between 5 and 5.5 kg/sq cm

Question 2.4. Proportion NOT between 3 and 7.5 kg/sq cm:

Mean (mu): 5 kg/sq cm
Standard Deviation (sigma): 1.5 kg/sq cm
Lower bound (x_lower): 3 kg/sq cm
Upper bound (x_upper): 7.5 kg/sq cm

Calculate the proportion between x_lower and x_upper
proportion_between = stats.norm.cdf(x_lower, loc=mu, scale=sigma)
minus stats.norm.cdf(x_lower, loc=mu, scale=sigma)

Calculate the proportion NOT between x_lower and x_upper
proportion_not_between = 1 - proportion_between

Question2.4 Answer:

Answer : Proportion of gunny bags with breaking strength NOT between 3 and 7.5 kg/sq cm: 0.1390

Visual Reprsentation:

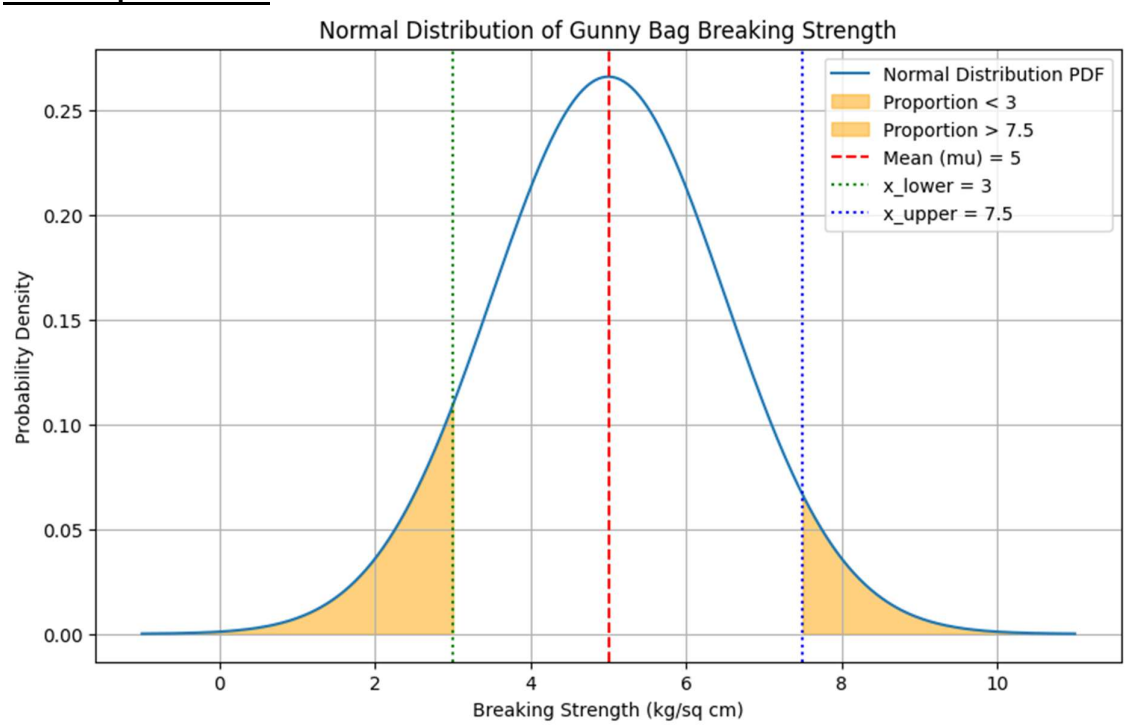

Figure 4: Proportion of gunny bags with breaking strength NOT between 3 and 7.5 kg/sq cm

6. Problem 3: Stone Printing Hardness Analysis

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

3.2 Is the mean hardness of the polished and unpolished stones the same?

6.1. Context and Objective

Zingaro stone printing requires a Brinell's hardness index of at least **150** for optimal printing. We analyze whether unpolished and polished stones meet these standards.

6.2. Data Overview & Analysis:

Table 2: Unpolished and Polished data

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

Figure 5: Head of data given

#	Column	Non-Null	Count	Dtype
0	Unpolished	75	non-null	float64
1	Treated and Polished	75	non-null	float64

index	Unpolished	Treated and Polished
std	33.0418044136061	15.587355273119856
min	48.40683839	107.5241671
count	75.0	75.0
25%	115.32975285	138.26829965000002
mean	134.11052653373332	147.78811718133335
50%	135.5971212	145.7213216
75%	158.215098	157.37331795
max	200.1613125	192.2728563

Figure 6: Table of Statistics Summary Unpolished vs Polished

- There are no null values.
- **Continuous Numerical Data:** Both the Unpolished and Treated and Polished columns contain float64 data types
- **Uniformity in Observations:** Both columns have exactly 75 non-null observations, indicating that the dataset is balanced and there are no missing values in either category.

6.3. Inferences

Mean:

- Unpolished Stones: The mean hardness index is approximately 134.11 kg per sq cm.
- Treated and Polished Stones: The mean hardness index is notably higher at approximately 147.79 kg per sq cm.

Standard Deviation:

- Unpolished Stones: Have a standard deviation of about 33.04, indicating a wider spread in hardness values.
- Treated and Polished Stones: Have a significantly lower standard deviation of about 15.59, suggesting that the hardness values are more consistent and less spread out.

Range, Min, Max Values:

- Unpolished Stones: The hardness ranges from a minimum of 48.41 to a maximum of 200.16. The 50th percentile (median) is 135.60, which is very close to the mean, suggesting a relatively symmetrical distribution.
- Treated and Polished Stones: The hardness ranges from a minimum of 107.52 to a maximum of 192.27. The median is 145.72, also close to the mean, indicating a relatively symmetrical distribution. The minimum hardness for treated stones is much higher than for unpolished stones, and the range is narrower.

6.4. Question 3.1. Suitability of Unpolished Stones:

To assess this, perform a one-sample t-test on the 'Unpolished' stones' hardness against the required minimum of 150.

6.4.1. Hypothesis:

Null Hypothesis (H0): **The true mean hardness of unpolished stones is greater than or equal to 150.**

H0: Mean hardness of unpolished stones >= 150

Alternative Hypothesis (H1): **The true mean hardness of unpolished stones is less than 150.**

H1: Mean hardness of unpolished stones < 150

6.4.2. Assumptions for all tests:

Level of significance is assumed as 5%

6.4.3. Values:

Hardness Threshold: 150

Significance Level: 0.05

T-statistic: -4.1646

P-value: 0.0000

Since the p-value (0.0000) is less than the significance level (0.05), we reject the null hypothesis.

The sample mean for unpolished stones is 134.11, which is significantly below 150.

6.4.4. Question3.1 Answer:

Answer: Zingaro is justified in thinking that the unpolished stones may not be suitable for printing, as their mean hardness is significantly less than 150.

6.4.5. Visual Representation:

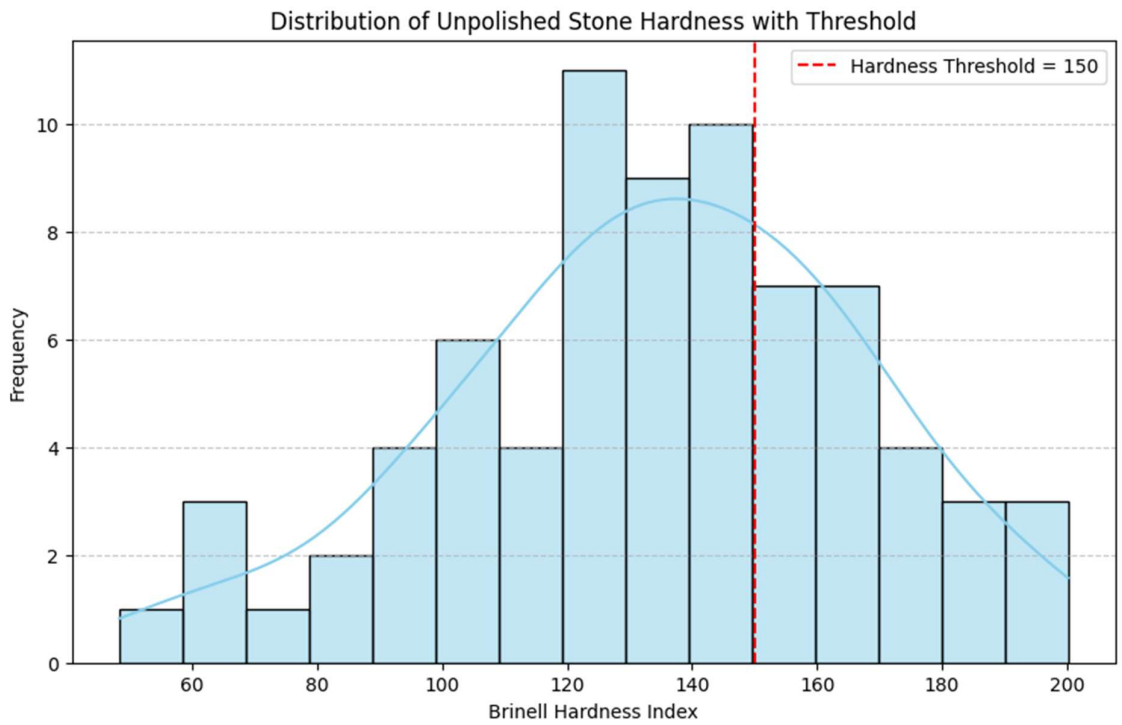

Figure 7: unpolished stones may not be suitable for printing

6.4.6. Conclusion:

Zingaro is justified; unpolished stones are generally unsuitable.

6.5. Question 3.2. Mean Hardness Comparison:

Is the mean hardness of the polished and unpolished stones the same?

To assess this, perform an independent two-sample t-test to compare the means of the two groups (unpolished and polished stones).

6.5.1. Hypothesis:

Null Hypothesis (H0): The mean hardness of unpolished stones is equal to the mean hardness of polished stones.

H0: Mean hardness of unpolished stones = Mean hardness of polished stones

Alternative Hypothesis (H1): The mean hardness of unpolished stones is not equal to the mean hardness of polished stones.

H1: Mean hardness of unpolished stones != Mean hardness of polished stones

6.5.2. Assumptions for all tests:

Level of significance is assumed as 5%

6.5.3. Testing Hypothesis:

This is a two-tailed test. A **Two-Sample T-test** was conducted to compare polished vs. unpolished stones.

Significance Level: 0.05

T-statistic: -3.2422

P-value: 0.0015

6.5.4. Result of Testing Hypothesis:

Since the p-value (0.0015) is less than the significance level (0.05), we reject the null hypothesis.

Question 3.2 Answer: The mean hardness of unpolished stones is significantly different from the mean hardness of polished stones.

6.5.5. Visual Representation

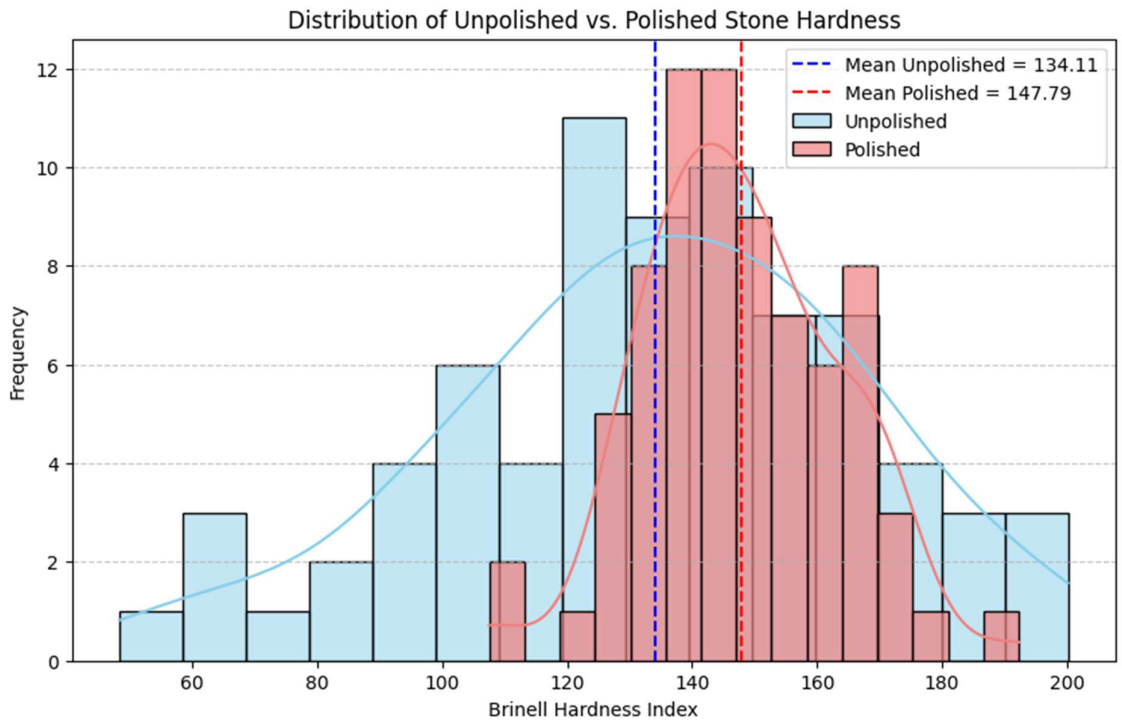

Figure 8: mean hardness of the polished and unpolished stones is significantly different

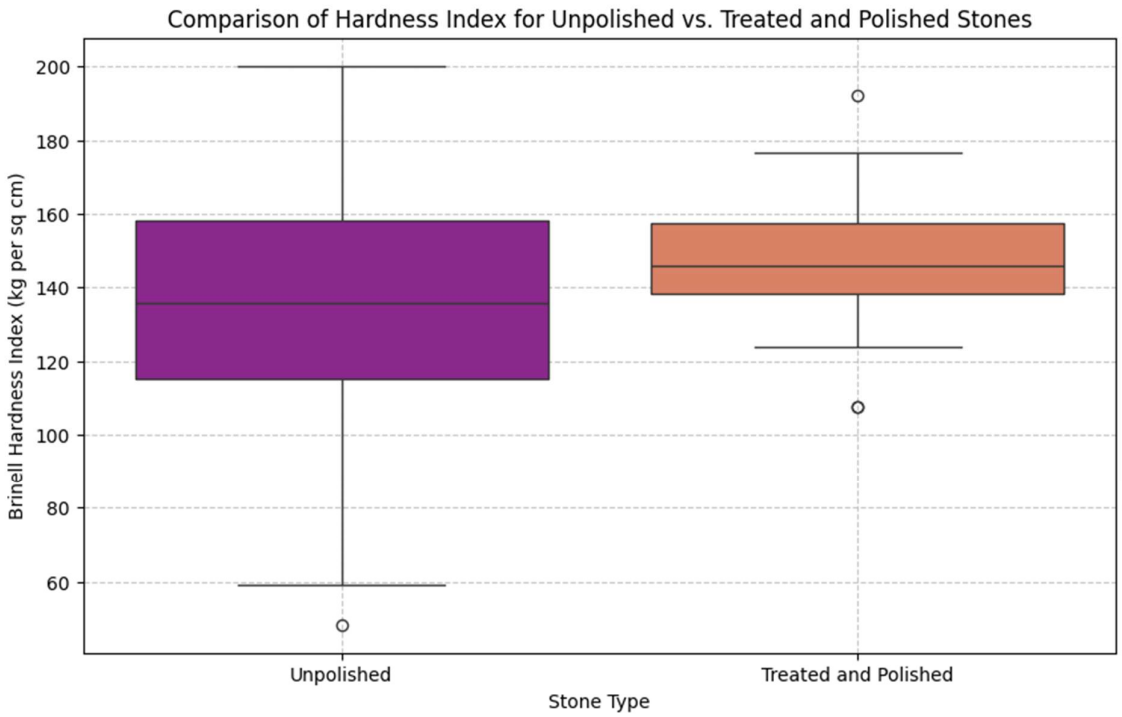

Figure 9:Hardness Comparison BoxPlot

6.5.6. Conclusion:

Since $p < 0.05$, we reject H_0. The mean hardness of polished stones (147.79) is significantly different from unpolished stones (134.11). Zingaro therefore has reason to believe that unpolished stones may not be suitable for printing, and this belief is justified.

---

7. Problem 4: Dental Implant Hardness

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest.

4.1 How does the hardness of implants vary depending on dentists?

4.2 How does the hardness of implants vary depending on methods?

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

4.4 How does the hardness of implants vary depending on dentists and methods together?

7.1. Context and Analysis Objective

This study examines how the hardness of dental implants is affected by three factors: Dentist, Method, and Alloy.

7.2. Data Overview and Analysing Data :-

	Dentist	Method	Alloy	Temp	Response
0	1	1	1	1500	813
1	1	1	1	1600	792
2	1	1	1	1700	792
3	1	1	2	1500	907
4	1	1	2	1600	792

Figure 10: Given dentist data

#	Column	Non-Null	Count	Dtype
0	Dentist	90	non-null	int64
1	Method	90	non-null	int64
2	Alloy	90	non-null	int64
3	Temp	90	non-null	int64
4	Response	90	non-null	int64

There are no null values.

	Dentist	Method	Alloy	Temp	Response
count	90.000000	90.000000	90.000000	90.000000	90.000000
mean	3.000000	2.000000	1.500000	1600.000000	741.777778

	Dentist	Method	Alloy	Temp	Response
std	1.422136	0.821071	0.502801	82.107083	145.767845
min	1.000000	1.000000	1.000000	1500.000000	289.000000
25%	2.000000	1.000000	1.000000	1500.000000	698.000000
50%	3.000000	2.000000	1.500000	1600.000000	767.000000
75%	4.000000	3.000000	2.000000	1700.000000	824.000000
max	5.000000	3.000000	2.000000	1700.000000	1115.000000

Figure 11:Statistical Summary of Dental data

The dataset has 90 rows and 5 columns. The dataset encompasses categorical and product-related information for Dental implant Materials. Columns Dentist, Method, Alloy, and Temp (Temperature) contain categorical data (Independent factors), whereas Response has the numerical data (Dependent factor)

Type of Data:

- The columns Dentist, Method, Alloy, and Temp (Temperature) represent categorical factors rather than continuous measurements. Despite being stored as integers (e.g., 1, 2, 3), these are nominal or ordinal categories.
- The columns Dentist, Method, and Alloy should likely be changed from Integer to Object (Category) to ensure the model treats them as groups rather than numerical scales.
- No missing values, indicating a balanced dataset for analysis
- Change the data type of Dentist, Method, Alloy, and Temp Columns to Object

```
#   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Dentist   90 non-null     object
 1   Method    90 non-null     object
 2   Alloy     90 non-null     object
 3   Temp      90 non-null     object
 4   Response  90 non-null     int64
```

Table 3 Dentist Table

	Dentist	Method	Alloy	Temp	Response
count	90.0	90.0	90.0	90.0	90.000000
unique	5.0	3.0	2.0	3.0	NaN
top	1.0	1.0	1.0	1500.0	NaN
freq	18.0	30.0	45.0	30.0	NaN
mean	NaN	NaN	NaN	NaN	741.777778
std	NaN	NaN	NaN	NaN	145.767845
min	NaN	NaN	NaN	NaN	289.000000
25%	NaN	NaN	NaN	NaN	698.000000
50%	NaN	NaN	NaN	NaN	767.000000
75%	NaN	NaN	NaN	NaN	824.000000
max	NaN	NaN	NaN	NaN	1115.000000

7.3. Key Analysis

7.4. Question 4.1. Variation by Dentist:

4.1 How does the hardness of implants vary depending on dentists?

Perform a One-Way ANOVA test. This will help us determine if there's a statistically significant difference in mean hardness across the different dentists.

7.4.1. Hypothesis:

Null Hypothesis (H0): The mean hardness of implants is the same across all dentists.

Alternative Hypothesis (H1): At least one dentist's mean hardness is different from the others.

7.4.2. Assumptions of Test:

The assumptions of the One-Way ANOVA test, to determine how the hardness of implants varies depending on dentists, are as follows:

Independence of Observations: The observations within each group (dentist) must be independent of each other, and the observations between groups must also be independent.

Normality: The dependent variable (implant hardness/Response) should be approximately normally distributed for each group (each dentist). ANOVA is relatively robust to minor deviations from normality, especially with larger sample sizes.

Homogeneity of Variances: The variance of the dependent variable (implant hardness) should be approximately equal across all groups (all dentists). This is often tested using Levene's test.

7.4.3. Statistical Analysis:

	mean	**median**	**std**	**min**	**max**
Dentist					
1	783.055556	792.0	129.281486	560	1115
2	786.666667	787.0	77.431564	673	933
3	748.611111	743.0	69.725629	613	858
4	713.666667	767.0	178.209064	289	894
5	676.888889	711.0	206.468881	312	1048

7.4.4. Shapiro-Wilk's test :

```
Dentist 1: Statistics=0.929, p=0.186
 -> Sample looks Gaussian (fail to reject H0 for normality)
Dentist 2: Statistics=0.967, p=0.731
 -> Sample looks Gaussian (fail to reject H0 for normality)
Dentist 3: Statistics=0.957, p=0.552
 -> Sample looks Gaussian (fail to reject H0 for normality)
Dentist 4: Statistics=0.809, p=0.002
 -> Sample does not look Gaussian (reject H0 for normality)
Dentist 5: Statistics=0.955, p=0.502
 -> Sample looks Gaussian (fail to reject H0 for normality)
```

Dentist 4's data significantly deviates from normality. The p-value is less than 0.05, leading us to reject the null hypothesis of normality for this group.

All other Dentist's show's normal distribution.The p-value is greater than 0.05, leading us to accept the null hypothesis of normality for this group.

However, ANOVA is generally robust to mild violations of normality. Hence we will progress with ANOVA.

7.4.5. Levene's test :

```
Statistics=3.708, p=0.008   -> Variances are not equal (reject H0 for equal variances)
```

Since the p-value (0.008) is less than the significance level of 0.05, we reject the null hypothesis that the variances are equal across all dentist groups. This means the assumption of homogeneity of variances is violated.

Combining this with our previous finding that Dentist 4's data did not look Gaussian, the assumptions for a traditional One-Way ANOVA are not fully met. In such cases, one might consider robust ANOVA methods (Welch's ANOVA) or data transformations. However, since the problem requested to determine how hardness varies depending on dentists using ANOVA, we'll proceed with the interpretation, keeping these violations in mind

7.4.6. ANOVA Test :

F-statistic: 1.9345

P-value: 0.1121

Since the p-value (0.1121) is greater than the significance level (0.05), we fail to reject the null hypothesis.

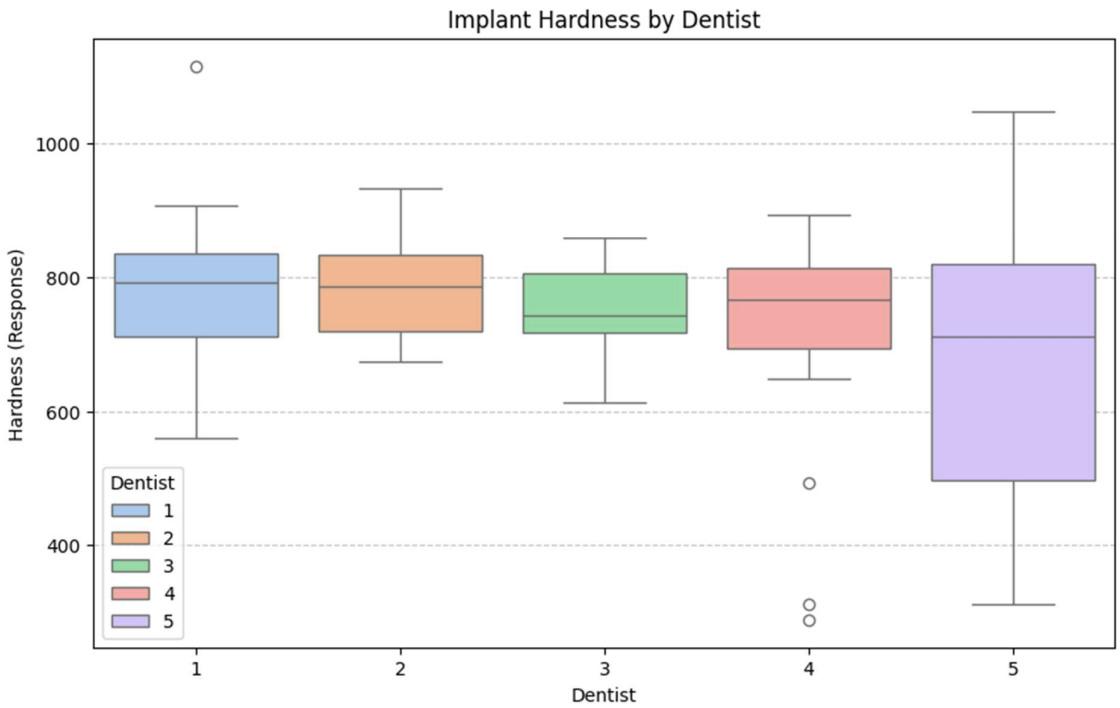

Figure 12 BoxPlot No difference in implant hardness depending on dentists

- **Mean and Median:** The box plot visually confirms the differences in median (the line inside the box) and interquartile range (the box itself) across dentists. Dentists 1 and 2 appear to have higher median hardness, aligning with their higher mean values. Dentist 5's box is positioned lower, indicating generally lower hardness.
- **Variability and Standard Deviation**: The length of the boxes (IQR) and the whiskers in the box plot correspond to the standard deviations. Dentist 5's box and whiskers are visibly longer, confirming its higher variability. Dentist 3 has a more compact box, reflecting its lower standard deviation.
- **Outliers**: The box plot clearly identifies potential outliers (individual points beyond the whiskers). For instance, Dentist 4 shows several lower outliers, which explains the discrepancy between its mean and median and contributes to its higher standard deviation.

Conclusion: There is no statistically significant difference in implant hardness depending on dentists.

7.5. Question 4.2 How does the hardness of implants vary depending on methods?

Perform a One-Way ANOVA test. This will help us determine if there's a statistically significant difference in mean hardness across the different methods.

7.5.1. Hypothesis:

Null Hypothesis (H0): The mean hardness of implants is the same across all methods.

Alternative Hypothesis (H1): At least one method's mean hardness is different from the others.

7.5.2. Assumptions of Test:

The assumptions for the One-Way ANOVA test are as follows:

Independence of Observations: The observations within each group (method) must be independent of each other, and the observations between groups must also be independent.

Normality: The dependent variable (implant hardness/Response) should be approximately normally distributed for each group (each method). ANOVA is relatively robust to minor deviations from normality, especially with larger sample sizes.

Homogeneity of Variances: The variance of the dependent variable (implant hardness) should be approximately equal across all groups (all methods). This is often tested using Levene's test or Bartlett's test.

7.5.3. Shapiro-Wilk's test :

Method 1: Statistics=0.965, p=0.418

-> Sample looks Gaussian (fail to reject H0 for normality)

Method 2: Statistics=0.937, p=0.076

-> Sample looks Gaussian (fail to reject H0 for normality)

Method 3: Statistics=0.909, p=0.014

-> Sample does not look Gaussian (reject H0 for normality)

While Method 1 and 2 show normal distribution, Method 3's data significantly deviates from normality. This is an important consideration for the ANOVA test, as it might affect the reliability of the results if not addressed.

However, ANOVA is generally robust to mild violations of normality, especially with larger sample sizes.

7.5.4. Levene's test :

Statistics=5.849, p=0.004

-> Variances are not equal (reject H0 for equal variances)

:Since the p-value (0.004) is less than the significance level of 0.05, we reject the null hypothesis that the variances are equal across all method groups. This means the assumption of homogeneity of variances is violated.

In such cases, robust ANOVA methods (like Welch's ANOVA) or data transformations might be considered for a more rigorous analysis.

However, for this problem, we will proceed with the interpretation of the One-Way ANOVA results, keeping these violations in mind

7.5.5. ANOVA Test Values:

F-statistic: 19.8927

P-value: 0.0000

Since the p-value (0.0000) is less than the significance level (0.05), we reject the null hypothesis.

Conclusion: There is a statistically significant difference in implant hardness depending on methods.

7.5.6. Statistical Inferences:

Table 4 statistical Analysis

Method	mean	median	std	min	max
1	793.9	792.0	86.4727782559942	627	1048
2	804.3333333333334	808.0	95.77103922388535	613	1115
3	627.1	681.0	167.62321253129096	289	847

Initial observations from the descriptive statistics showed a notable difference in mean hardness across methods:

Methods 1 and 2 had similar higher mean hardness, while Method 3 had a significantly lower mean and higher standard deviation, suggesting greater variability.

7.5.7. Visual Representation:

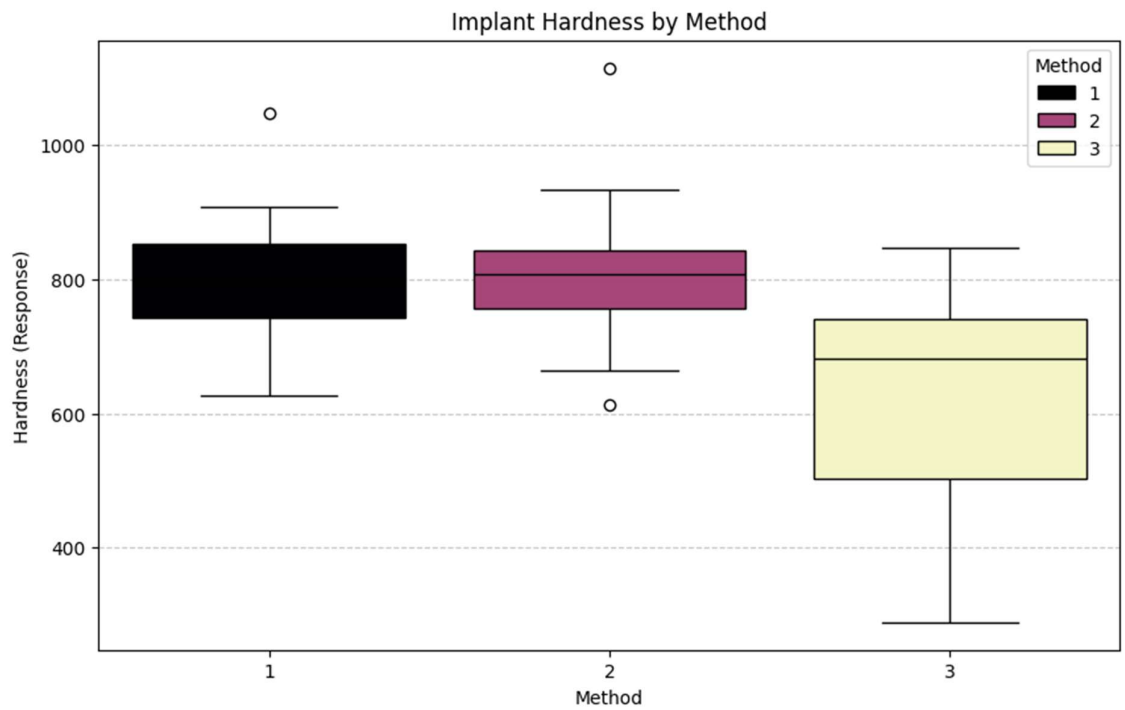

Figure 13 BoxPlot statistically significant difference in implant hardness depending on methods

The visual differences observed in the box plot for methods strongly supported this statistical finding.

This means that the **method used for dental implants does indeed have a significant impact on their hardness**.

Conclusion: There is a statistically significant difference in implant hardness depending on methods.

7.6. Question 4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

To check the interaction effect between 'Dentist' and 'Method' on the hardness of dental implants for each type of alloy, we will prepare interaction plot for Each Alloy.

7.6.1. Visual Representation for Alloy1:

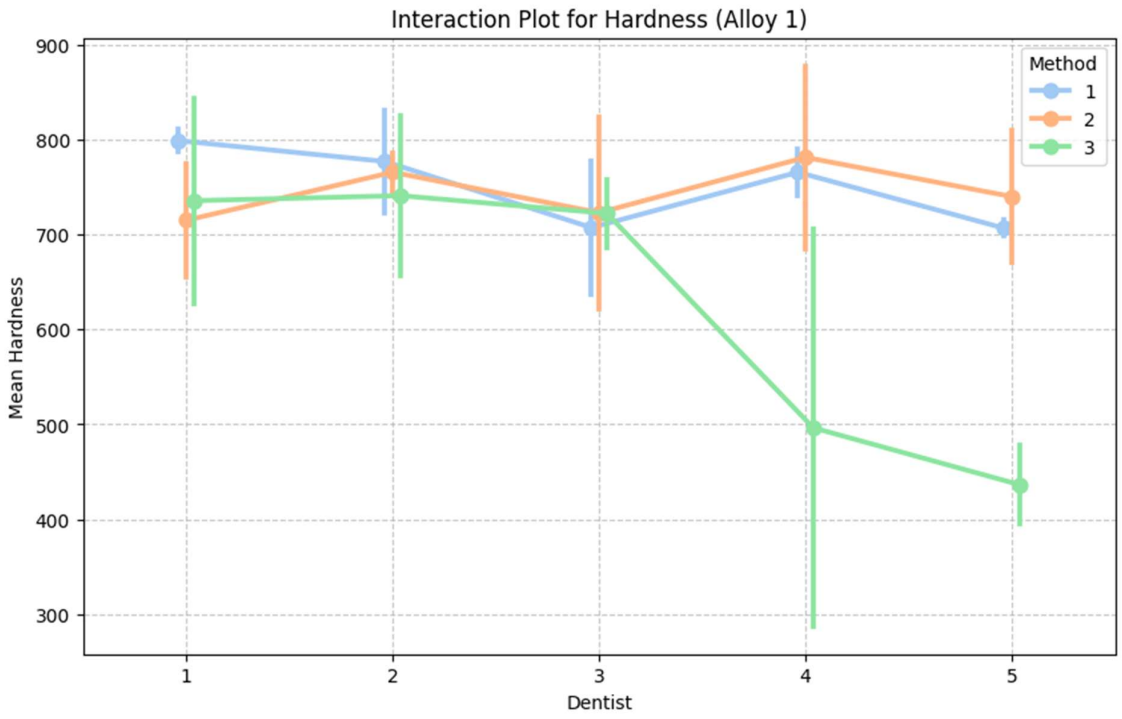

Figure 14 Interaction Plot for Alloy1

Insights:

- The interaction plot for Alloy 1 shows non-parallel lines connecting the mean hardness values for different methods across the various dentists.
- This non-parallelism indicates that the effect of the 'Method' on implant 'Hardness' depends on which 'Dentist' is performing the procedure, and vice versa. This suggests that certain dentist-method combinations lead to unique hardness outcomes.

7.6.2. Visual Representation for Alloy2:

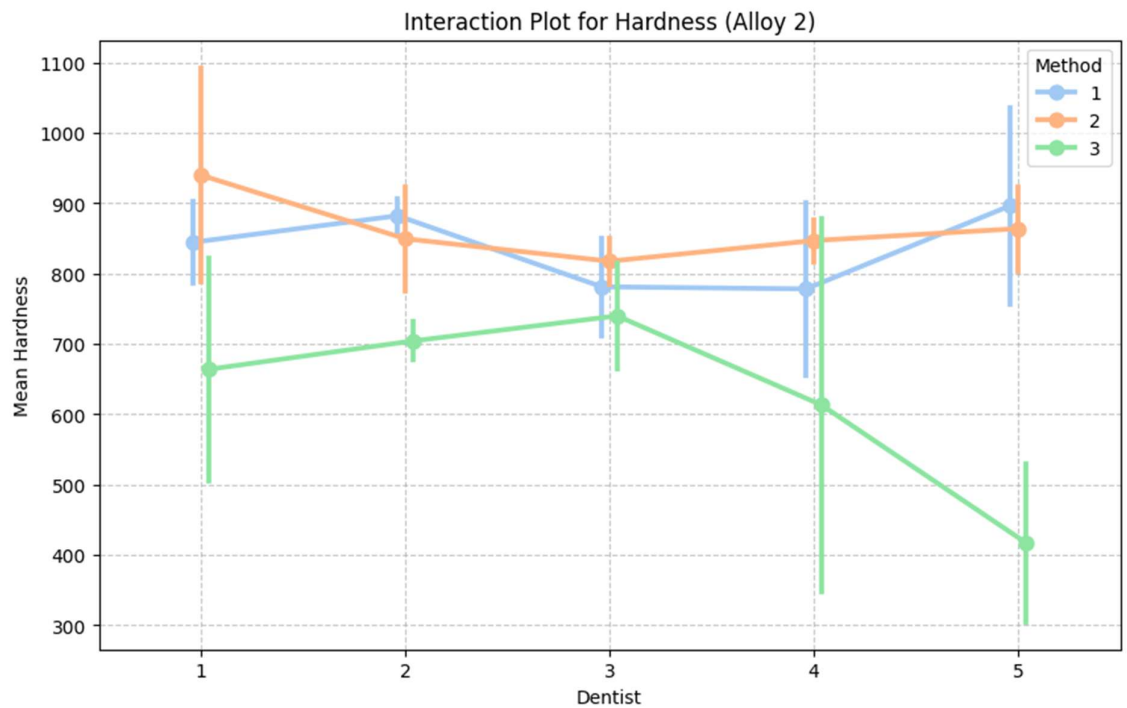

Figure 15 Interaction Plot for Alloy2

Insights:

- The interaction plot for Alloy 2 (displayed previously) shows lines that are relatively parallel to each other, although there might be minor visual deviations.
- The near-parallel lines suggest that the effect of the 'Method' on implant 'Hardness' is generally consistent across different 'Dentists' for Alloy 2. In other words, if Method 1 yields higher hardness than Method 3, this relationship tends to hold true regardless of which dentist performs the procedure.

7.6.3. Conclusion for Question 4.3:

- For Alloy Type 1, the choice of method significantly interacts with the dentist's performance, meaning that the optimal method might depend on the specific dentist.
- For Alloy Type 2, the impact of different methods on implant hardness is more consistent regardless of the dentist, suggesting that the methods themselves have a more uniform effect that isn't heavily influenced by the individual dentist.

7.7. Question 4.4 How does the hardness of implants vary depending on dentists and methods together?

For each unique 'Alloy' type, the data will be filtered. Then, a two-way ANOVA will be performed using statsmodels to analyze the effect of 'Dentist', 'Method', and their interaction on 'Response' (hardness).

The p-values from the ANOVA for the interaction term (Dentist:Method) will be examined for each alloy to determine if a statistically significant interaction effect exists at a 5% significance level.

7.7.1. FORMULA for ANOVA

Alloy Type: 1 :

ANOVA Model Formula: Response ~ C(Dentist) + C(Method) + C(Dentist):C(Method)

Alloy Type: 2

ANOVA Model Formula: Response ~ C(Dentist) + C(Method) + C(Dentist):C(Method)

7.7.2. Hypothesis:

Null Hypothesis (H0):

There is no interaction effect between Dentist and Method on the mean hardness of dental implants.

Alternative Hypothesis (H1): There is a statistically significant interaction effect between Dentist and Method on the mean hardness of dental implants.

7.7.3. Assumptions of Test

The assumptions for the two-way ANOVA test, which was used in section 4.3 to examine the interaction effect between 'Dentist' and 'Method' on implant hardness for each alloy type, are as follows:

1. Independence of Observations: All observations within and between groups must be independent.
2. Normality: The dependent variable (implant hardness/Response) should be approximately normally distributed for each combination of the independent variables (each Dentist-Method combination within each Alloy type). ANOVA is generally robust to moderate violations of normality, especially with larger sample sizes.
3. Homogeneity of Variances (Homoscedasticity): The variance of the dependent variable should be approximately equal across all groups (i.e., for all combinations of Dentist and Method within each Alloy type). Levene's test is commonly used to check this assumption.
4. Categorical Independent Variables: The independent variables ('Dentist' and 'Method') must be categorical (nominal or ordinal).
5. Continuous Dependent Variable: The dependent variable ('Response' / hardness) must be continuous and measured on an interval or ratio scale.

7.7.4. Shapiro-Wilk Test

```
--- Analyzing for Alloy 1 ---

Shapiro-Wilk Test for Normality for Alloy 1 (by Dentist-Method Group):

  Not enough data points for Dentist 1, Method 1. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 1, Method 2. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 1, Method 3. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 2, Method 1. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 2, Method 2. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 2, Method 3. Skipping Shapiro-Wilk test

  Not enough data points for Dentist 3, Method 1. Skipping Shapiro-Wilk test
```

```
Not enough data points for Dentist 3, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 3, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 3. Skipping Shapiro-Wilk test
```

```
--- Analyzing for Alloy 2 ---
Shapiro-Wilk Test for Normality for Alloy 2 (by Dentist-Method Group):
Not enough data points for Dentist 1, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 1, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 1, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 2, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 2, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 2, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 3, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 3, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 3, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 4, Method 3. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 1. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 2. Skipping Shapiro-Wilk test
Not enough data points for Dentist 5, Method 3. Skipping Shapiro-Wilk test
```

- **Alloy1 - Shapiro-Wilk Test** for Normality: Could not perform because most (Dentist-Method) subgroups have 3 or fewer points, for which the Shapiro-Wilk test is not typically applicable or reliable.

- **Alloy2 - Shapiro-Wilk Test** for Normality: Similar to Alloy 1, could not perform because most (Dentist-Method) subgroups have 3 or fewer points, for which the Shapiro-Wilk test is not typically applicable or reliable

7.7.5. Levene's test :

Analyzing for Alloy 1:
Statistics=1.219, p=0.313 -> Variances are equal (Fail to reject H0 for equal variances)
Analyzing for Alloy 2
Statistics=0.671, p=0.783 -> Variances are equal (Fail to reject H0 for equal variances)

Alloy1:

Homogeneity of Variances: The p-value is 0.313. Since this is greater than 0.05, we fail to reject the null hypothesis, meaning that the variances are considered equal across the Dentist-Method groups for Alloy 1.

Alloy2:

Homogeneity of Variances: The p-value is 0.783. Since this is greater than 0.05, we fail to reject the null hypothesis, meaning that the variances are considered equal across the Dentist-Method groups for Alloy 2.

For both alloys, the assumption of homogeneity of variances is met, which is a good indication for proceeding with ANOVA

7.7.6. ANOVA test :

A two-way ANOVA will be performed using statsmodels to analyze the effect of 'Dentist', 'Method', and their interaction on 'Response' (hardness).

The p-values from the ANOVA for the interaction term (Dentist:Method) will be examined for each alloy to determine if a statistically significant interaction effect exists at a 5% significance level.

<u>Alloy Type 1: ANOVA Results for C(Dentist):C(Method) (Interaction Term):</u>

- F-statistic: 3.398
- P-value: 0.0068

- **Interpretation:** Since the p-value (0.0068) for the interaction term is less than the significance level (0.05), we **reject the null hypothesis**. This indicates that there is a **statistically significant interaction effect** between the 'Dentist' and 'Method' on the hardness of dental implants for Alloy Type 1. The effect of the dental method on implant hardness depends on the specific dentist performing the procedure, and vice-versa.
- **Visual Observation (Interaction Plot for Alloy Type 1):** The lines representing different methods are not parallel, indicating that the impact of a particular method on hardness changes depending on the dentist. For example, some dentists might perform better with one method, while others excel with a different one, suggesting a crossover or varying trends.

<u>Alloy Type 2: ANOVA Results for C(Dentist):C(Method) (Interaction Term):</u>

- F-statistic: 1.9228
- P-value: 0.0932

<u>Overall : Alloy Type 1:</u>

- **Main Effect of Dentist (C(Dentist))**: The p-value is 0.0115. Since 0.0115 < 0.05, we reject the null hypothesis. This indicates that there is a statistically significant difference in the mean hardness of implants across different dentists for Alloy Type 1.
- **Main Effect of Method (C(Method))**: The p-value is 0.0003. Since 0.0003 < 0.05, we reject the null hypothesis. This indicates that there is a statistically significant difference in the mean hardness of implants across different methods for Alloy Type 1.
- **Interaction Effect (C(Dentist):C(Method))**: The p-value is 0.0068. Since 0.0068 < 0.05, we reject the null hypothesis. This suggests that the effect of the 'Method' on implant hardness depends on the 'Dentist' (and vice-versa) for Alloy Type 1. The interaction plot for Alloy Type 1 visually confirms this, as the lines representing different methods are not parallel across dentists, indicating an interaction.

<u>Overall Alloy Type 2:</u>

- **Main Effect of Dentist (C(Dentist))**: The p-value is 0.3718. Since 0.3718 > 0.05, we fail to reject the null hypothesis. This indicates that there is no statistically significant difference in the mean hardness of implants across different dentists for Alloy Type 2.
- **Main Effect of Method (C(Method))**: The p-value is 0.000004. Since 0.000004 < 0.05, we reject the null hypothesis. This indicates that there is a statistically significant difference in the mean hardness of implants across different methods for Alloy Type 2.

- **Interaction Effect (C(Dentist):C(Method))**: The p-value is 0.0932. Since 0.0932 > 0.05, we fail to reject the null hypothesis. This suggests that there is no statistically significant interaction effect between 'Dentist' and 'Method' on implant hardness for Alloy Type 2. The lines in the interaction plot for Alloy Type 2 appear to be more parallel compared to Alloy Type 1, supporting the lack of significant interaction.

7.7.7. Findings:

The hardness of implants varies depending on dentists and methods, but the nature of this variation is influenced by the type of alloy:

For Alloy Type 1: Both the dentist and the method individually have a significant impact on implant hardness. Furthermore, there is a significant interaction effect, meaning the effectiveness of a particular method on hardness depends on which dentist is performing it. This implies that for Alloy Type 1, dentists may achieve different hardness results depending on the method they use, and some dentist-method combinations might be more effective than others.

For Alloy Type 2: The method used significantly influences implant hardness, but the individual dentist does not have a statistically significant effect. Crucially, there is no significant interaction effect between dentist and method. This suggests that for Alloy Type 2, while the choice of method is important for hardness, all dentists tend to achieve similar results for a given method, and the impact of the method on hardness is consistent regardless of the dentist.

7.7.8. Visual Representation for Alloy1:

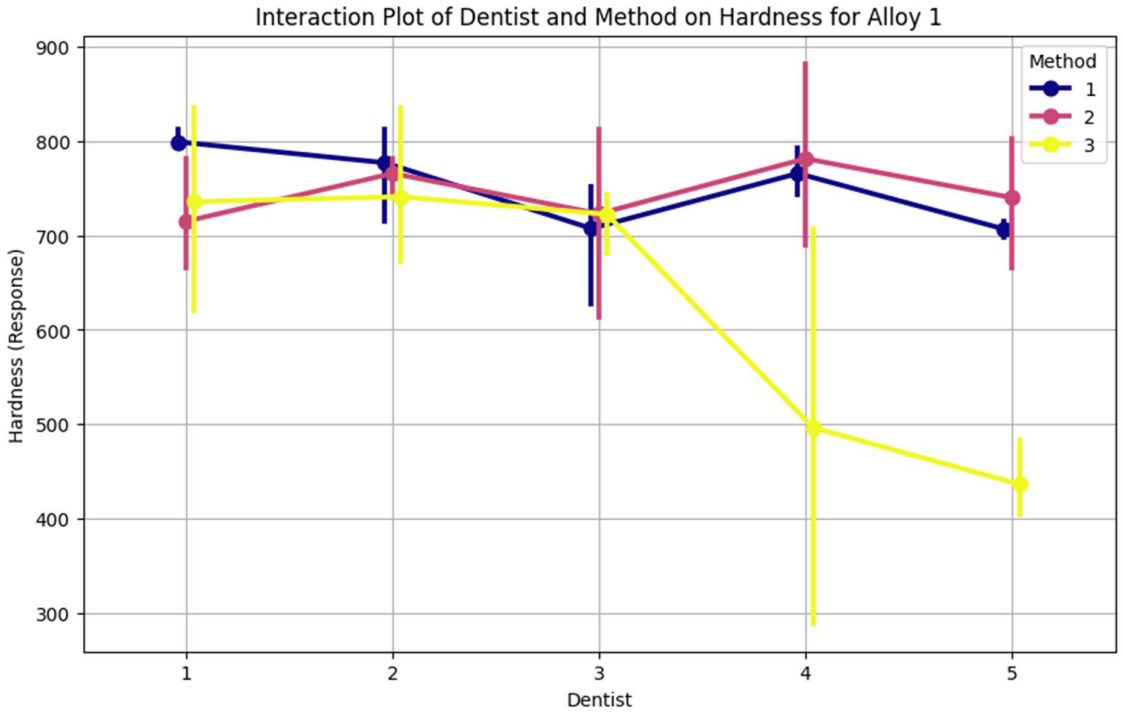

Figure 16 interaction effect between 'Dentist' and 'Method' for Alloy1

Based on the ANOVA results and the interaction plots from the previous analysis for each alloy type, here's how the hardness of implants varies depending on dentists and methods together:

<u>Interpretation of ANOVA Results (Significance Level = 0.05):</u>

Alloy Type 1: Main Effect of Dentist: The p-value is 0.0115. Since this is less than 0.05, we reject the null hypothesis, indicating a statistically significant difference in the mean hardness of implants across different dentists for Alloy Type 1.

Main Effect of Method: The p-value is 0.0003. Since this is less than 0.05, we reject the null hypothesis, indicating a statistically significant difference in the mean hardness of implants across different methods for Alloy Type 1.

Interaction Effect (Dentist:Method): The p-value is 0.0068. Since this is less than 0.05, we reject the null hypothesis, suggesting that the effect of the 'Method' on implant hardness depends on the 'Dentist' (and vice-versa) for Alloy Type 1. The interaction plot visually confirms this, as the lines representing different methods are not parallel across dentists, indicating an interaction.

7.7.9. Visual Representation for Alloy2:

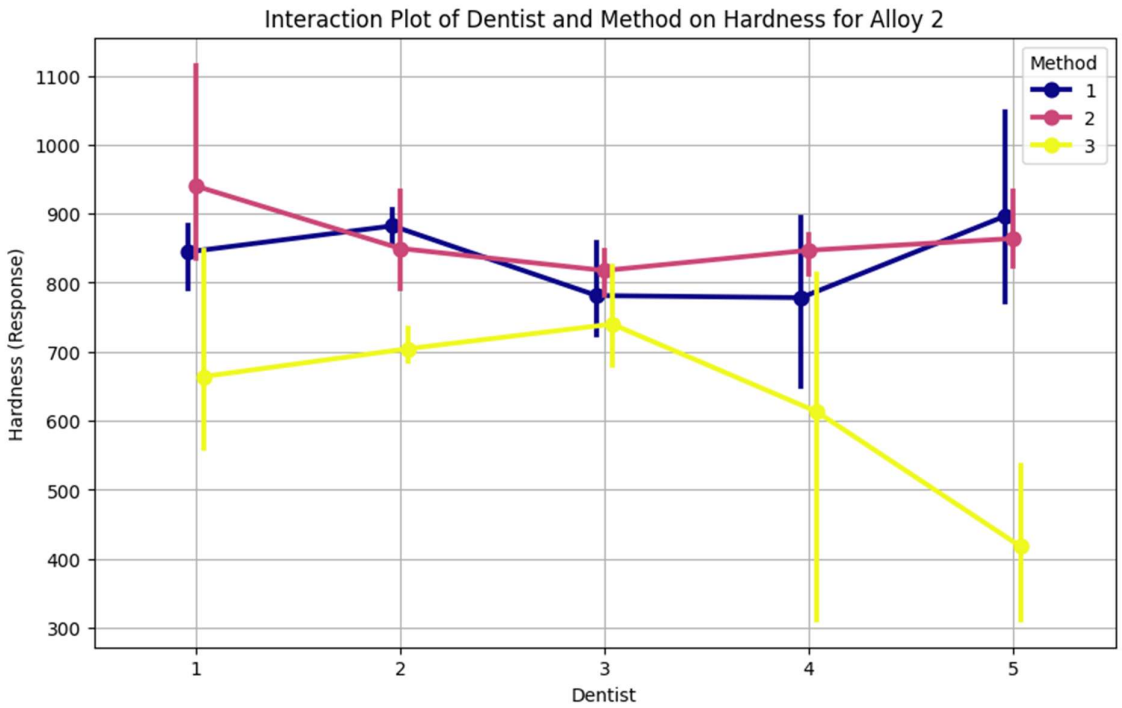

Figure 17 interaction effect between 'Dentist' and 'Method' for Alloy 2

Alloy Type 2: Main Effect of Dentist: The p-value is 0.3718. Since this is greater than 0.05, we fail to reject the null hypothesis, indicating no statistically significant difference in the mean hardness of implants across different dentists for Alloy Type 2.

Main Effect of Method: The p-value is 0.000004. Since this is less than 0.05, we reject the null hypothesis, indicating a statistically significant difference in the mean hardness of implants across different methods for Alloy Type 2.

Interaction Effect (Dentist:Method): The p-value is 0.0932. Since this is greater than 0.05, we fail to reject the null hypothesis, suggesting no statistically significant interaction effect between 'Dentist' and 'Method' on implant hardness for Alloy Type 2. The lines in the interaction plot for Alloy Type 2 appear to be more parallel, supporting the lack of significant interaction.

7.7.10. Summary

The analysis addresses how implant hardness varies depending on 'Dentist' and 'Method' together, including their main and interaction effects for each alloy type.

- **For Alloy Type 1**: Implant hardness is significantly affected by the individual 'Dentist' (p-value = 0.0115), the 'Method' used (p-value = 0.0003), and there is a statistically significant interaction effect between 'Dentist' and 'Method' (p-value = 0.0068). This means the effectiveness of a particular method on hardness for Alloy Type 1 depends on which dentist is performing it.
- **For Alloy Type 2**: Implant hardness is significantly affected by the 'Method' used (p-value = 0.000004). However, the individual 'Dentist' does not have a statistically significant effect (p-value = 0.3718), and there is no statistically significant interaction effect between 'Dentist' and 'Method' (p-value = 0.0932). This implies that for Alloy Type 2, while the choice of method is crucial, its impact on hardness is consistent regardless of the dentist.

<u>Data Analysis Key Findings</u>

- For Alloy Type 1, a statistically significant interaction effect was found between 'Dentist' and 'Method' on implant hardness ($p = 0.0068$), indicating that the optimal method varies by dentist. Both the main effect of 'Dentist' ($p = 0.0115$) and 'Method' ($p = 0.0003$) were also significant.
- For Alloy Type 2, no statistically significant interaction effect was found between 'Dentist' and 'Method' on implant hardness ($p = 0.0932$). While the 'Method' alone significantly affected hardness ($p = 0.000004$), the 'Dentist' alone did not show a significant effect ($p = 0.3718$).
- Visualizations using interaction plots supported these findings, showing non-parallel lines for Alloy Type 1, suggesting interaction, and more parallel lines for Alloy Type 2, indicating less interaction.

<u>Insights or Next Steps</u>

- For Alloy Type 1, further investigation into specific dentist-method combinations that yield optimal hardness could be beneficial to develop best practices tailored to individual dentists.
- For Alloy Type 2, efforts should focus on optimizing the 'Method' itself, as its impact on implant hardness is consistent across different dentists. Training or standardization around the most effective methods would be impactful.

7.7.11. Question 4.4 Answer:

For Alloy Type 1: Both the dentist and the method individually have a significant impact on implant hardness. Furthermore, there is a significant interaction effect, meaning the effectiveness of a particular method on hardness depends on which dentist is performing it. This implies that for Alloy Type 1, dentists may achieve different hardness results depending on the method they use, and some dentist-method combinations might be more effective than others.

For Alloy Type 2: The method used significantly influences implant hardness, but the individual dentist does not have a statistically significant effect. Crucially, there is no significant interaction effect between dentist and method. This suggests that for Alloy Type 2, while the choice of method is important for hardness, all dentists tend to achieve similar results for a given method, and the impact of the method on hardness is consistent regardless of the dentist.

This means that for Alloy Type 1, you need to consider the specific combination of dentist and method, while for Alloy Type 2, focusing on optimizing the method itself is more crucial.

7.7.12. Summary Recommendations

- **For Alloy 1:** Standardize methods across dentists to reduce variability.
- **For Alloy 2:** Focus on optimizing the specific Method, as it is the primary driver of hardness regardless of who performs the procedure.

8. Overall Insights & Recommendations

Actionable insights and recommendations for each problem:

Problem 1 (Football Injuries):

- Implement targeted injury prevention programs for strikers and wingers, as these positions show a higher propensity for foot injuries.

Problem 2 (Gunny Bags Breaking Strength):

- Review the manufacturing process to improve consistency, as a significant proportion (11.12%) of bags fall below the critical breaking strength of 3.17 kg/sq cm, indicating potential quality control issues.

Problem 3 (Zingaro Stone Hardness):

- Zingaro should exclusively use polished stones or implement a pre-treatment to increase the hardness of unpolished stones to meet the minimum Brinell's hardness index of 150 required for optimal printing.

Problem 4 (Dental Implants Hardness):

- For **Alloy Type 1**, identify and standardize the specific dentist-method combinations that yield optimal hardness, as there is a significant interaction effect.
- For **Alloy Type 2**, focus on optimizing the methods themselves, as their impact on hardness is consistent across all dentists, indicating method is the primary driver.